

Over-the-counter ibuprofen and risk of gastrointestinal bleeding complications: a systematic literature review.

BACKGROUND: Exposure to over-the-counter (OTC) ibuprofen and other OTC non-steroidal anti-inflammatory drugs (NSAIDs) is substantial. Although the literature on gastrointestinal (GI) safety of NSAID therapy is extensive, the risk profiles of OTC and prescription dosing are seldom separated, and few studies provide risks specific to OTC ibuprofen. **OBJECTIVE:** To conduct a literature review to evaluate the risk of GI bleeding events related to OTC ibuprofen use. **METHODS:** Published clinical trials, observational studies, and meta-analyses of OTC ibuprofen use, defined as up to 1200 mg/day or stated as 'over the counter,' reporting endpoints of incidence rates and proportions of GI bleeding events (e.g., GI bleeding-related hospitalizations and deaths) were identified via MEDLINE through 2010. Data from these studies were summarized. **RESULTS:** Twenty studies (nine observational, ten clinical trials, one meta-analysis) reporting incidence rates and proportions of a GI bleeding-related event associated with OTC or OTC-specific doses of ibuprofen were included. The frequency of a GI-related hospitalization was <0.2% for patients on OTC-comparable doses. Incidence rates among those using OTC-comparable doses ranged from 0 to 3.19 per 1000 patient-years. The incidence of a GI bleeding-related event increased with age and the use of concomitant medications, and there was a general, though not always statistically significant, ibuprofen dose-response relationship. The relative risk of any GI bleeding-related event ranged from 1.1 to 2.4 for users of OTC-specific doses of ibuprofen compared to non-users. **CONCLUSIONS:** Studies reported low incidence of GI bleeding events with use of OTC ibuprofen. Few published studies that specifically investigated OTC ibuprofen use were identified. Varying methodologies and definitions of exposure and outcomes prevented direct comparison of many results. Only studies that used the methods herein described were identified. Further research evaluating the risk of GI bleeding events in patients taking OTC-specific ibuprofen use may be useful.